

AA-AMILZIDE

Hydrochlorothiazide and Amiloride Hydrochloride Tablets USP 50mg/5mg Diuretic - Antihypertensive

PHARMACOLOGY: AA-AMILZIDE (hydrochlorothiazide and amiloride hydrochloride) is a diuretic/antihypertensive combining the potent natriuretic action of hydrochlorothiazide with the potassium conserving property of amiloride hydrochloride. The mild diuretic and antihypertensive actions of amiloride hydrochloride are additive to the natriuretic, diuretic and antihypertensive activity of the thiazide while minimizing the loss of potassium and lessening the likelihood of acid base imbalance. The onset of the diuretic action of AA-AMILZIDE is within 1 to 2 hours and this action appears to be sustained for approximately 24 hours.

Hydrochlorothiazide: Hydrochlorothiazide is a diuretic and antihypertensive agent. It affects the renal tubular mechanism of electrolyte reabsorption. Hydrochlorothiazide increases excretion of sodium and chloride in approximately equivalent amounts and may cause a simultaneous, usually minimal, loss of bicarbonate. Natriuresis is usually accompanied by some loss of potassium. The mechanism of the antihypertensive effect of thiazides may be related to the excretion and redistribution of body sodium. Hydrochlorothiazide usually does not decrease normal blood pressure. The onset of the diuretic action of hydrochlorothiazide occurs in 2 hours and the peak action in about 4 hours. Diuretic activity lasts about 6 to 12 hours. Hydrochlorothiazide is eliminated rapidly by the kidney.

Non-melanoma skin cancer: Based on available data from epidemiological studies, cumulative dose-dependent association between hydrochlorothiazide and NMSC has been observed. One study included a population comprised of 71,533 cases of BCC and of 8,629 cases of SCC matched to 1,430,833 and 172,462 population controls, respectively. High hydrochlorothiazide use ($\geq 50,000$ mg cumulative) was associated with an adjusted OR of 1.29 (95% CI: 1.23-1.35) for BCC and 3.98 (95% CI: 3.68-4.31) for SCC. A clear cumulative dose response relationship was observed for both BCC and SCC. Another study showed a possible association between lip cancer (SCC) and exposure to hydrochlorothiazide: 633 cases of lip-cancer were matched with 63,067 population controls, using a risk-set sampling strategy. A cumulative dose-response relationship was demonstrated with an adjusted OR 2.1 (95% CI: 1.7-2.6) increasing to OR 3.9 (3.0-4.9) for high use ($\sim 25,000$ mg) and OR 7.7 (5.7-10.5) for the highest cumulative dose ($\sim 100,000$ mg).

Amiloride Hydrochloride: Amiloride hydrochloride is an antikaliuretic drug with mild natriuretic diuretic and antihypertensive activity. These activities may be additive to the effects of thiazides or other saluretic diuretic agents. The principal use of amiloride hydrochloride is to conserve potassium in selected patients receiving kaliuretic-diuretic agents. The action is not related to the level of aldosterone excretion. Amiloride hydrochloride is not an aldosterone antagonist. The drug acts directly on the distal portion of the nephron. Amiloride hydrochloride causes an increase in sodium excretion and a decrease in potassium and hydrogen ion excretion. Chloride excretion may remain unchanged or increase slowly with continued therapy. Approximately 50% of an oral dose is absorbed. Amiloride hydrochloride usually begins to act within 2 hours after an oral dose. Its effect on electrolyte excretion reaches a peak between 6 and 10 hours and lasts about 24 hours. Peak plasma levels are obtained in 3 to 4 hours and plasma half-life varies from 6 to 9 hours. Amiloride hydrochloride is not metabolized by the liver. About 50% of a 20mg dose amiloride hydrochloride is excreted unchanged in the urine and 40% is excreted in the stool within 72 hours. In clinical studies amiloride hydrochloride was found to have little effect on glomerular filtration rate or renal blood flow. Bioavailability studies were performed using normal human volunteers. The rate and extent of absorption after a single oral dose of 5mg amiloride and 50mg hydrochlorothiazide in the form of AA-AMILZIDE 5-50mg and Moduret 5-50mg was measured and compared. The results can be summarized as follows:

<u>Amiloride HCl</u>	<u>Moduret (SD*)</u>	<u>AA-Amilzide (SD*)</u>
AUC 0-24 (ng-hr/ml)	62.5 (21.0)	66.8 (18.4)
Cmax (ng/ml)	6.0 (2.2)	6.7 (2.4)
Tmax (hrs)	3.8 (0.6)	3.3 (0.8)
t ¹ / ₂ (hrs)	8.2 (2.5)	9.6 (3.8)

Hydrochlorothiazide

AUC 0-24 (ng-hr/ml)	1603 (414)	1654 (426)
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C _{max} (ng/ml)	235 (73)	242 (60)
T _{max} (hrs)	2.3 (1.0)	2.4 (0.8)
t _{1/2} (hrs)	9.1 (3.3)	10.0 (5.6)

*SD - Standard Deviation

INDICATIONS AND CLINICAL USES: Fixed-dose combination drugs are not indicated for initial therapy. Patients should be titrated on the individual drugs. If the fixed combination represents the dosage so determined, its use may be more convenient in patient management. If during maintenance therapy dosage adjustment is necessary, it is advisable to use the individual drugs.

AA-AMILZIDE (hydrochlorothiazide and amiloride hydrochloride) is indicated in the maintenance therapy of patients with hepatic cirrhosis with ascites and edema. Patients with edema of cardiac origin or with arterial hypertension who are hypokalemic or in whom maintenance of normal potassium levels is considered to be clinically important i.e., digitalized patients, patients in whom adequate dietary intake of potassium is not feasible or patients with cardiac arrhythmias. Use in Hepatic Cirrhosis with Ascites and Edema: Amiloride hydrochloride used alone may provide satisfactory diuresis with diminished potassium loss and with a reduced risk of metabolic alkalosis. In resistant cases amiloride hydrochloride may be used with kaliuretic diuretic agents to help produce satisfactory diuretics, while maintaining a more balanced serum electrolyte pattern. As with all therapy for the ascites of hepatic cirrhosis, gradual weight loss and avoidance of electrolyte imbalance are the chief objectives (see PRECAUTIONS).

CONTRAINDICATIONS: Hyperkalemia: AA-AMILZIDE (hydrochlorothiazide and amiloride hydrochloride) should not be used in the presence of elevated serum potassium levels (see WARNINGS AND PRECAUTIONS). Antikaliuretic Therapy or Potassium Salts: Other antikaliuretic agents and potassium supplements are contraindicated in patients receiving AA-AMILZIDE (such combination therapy is commonly associated with rapid increases in plasma potassium levels). Impaired Renal Function: Anuria, acute renal failure, severe or progressive renal disease and diabetic nephropathy are contraindications to the use of AA-AMILZIDE (see WARNINGS AND PRECAUTIONS). Hypersensitivity: AA-AMILZIDE is contraindicated in patients who are hypersensitive to either component or to other sulfonamide-derived drugs.

WARNINGS AND PRECAUTIONS:

Serious Warnings and Precautions:

Hyperkalemia: Hyperkalemia, i.e. serum potassium levels over 5.5mEq per litre, has been observed in some patients who received amiloride hydrochloride either alone or with diuretics. This has been noted particularly in elderly patients, in diabetic patients, and in hospitalized patients with hepatic cirrhosis or cardiac edema who had known renal impairment, were seriously ill, or were receiving vigorous diuretic therapy. Since fatalities have occurred in such patients, they should be monitored carefully for clinical, laboratory and electrocardiographic (ECG) evidence of hyperkalemia and for acidosis. Monitoring of the serum potassium level is important because hyperkalemia is not always associated with an abnormal ECG. Warning signs or symptoms of hyperkalemia include paresthesias, muscular weakness, fatigue, flaccid paralysis of the extremities, bradycardia, shock, and ECG abnormalities. When abnormal, the ECG in hyperkalemia is characterized primarily by tall, peaked T waves or elevations from previous tracings. There may also be lowering of the R wave and increased depth of the S wave, widening and even disappearance of the P wave, progressive widening of the QRS complex, prolongation of the PR interval and ST depression. If the serum potassium level exceeds 6.5 mEq per litre, active measures should be taken to reduce it. Such measures include the intravenous administration of sodium bicarbonate solution or oral or parenteral glucose with rapid-acting insulin preparation. If needed, a cation exchange resin such as sodium polystyrene sulfonate may be given orally or by enema. Patients with persistent hyperkalemia may require dialysis.

Electrolyte Imbalance and BUN Increases: Hyponatremia and hypochloremia may occur during the use of AA-AMILZIDE. Hypokalemia may also occur although the incidence is less than with thiazides alone. Any chloride deficit is usually mild and may be corrected by the use of ammonium chloride (except in patients with hepatic disease) and largely prevented by a near normal salt intake. Increases in BUN levels have been reported and have usually accompanied vigorous fluid elimination, especially when diuretic combinations were used in seriously ill patients, such as those who have hepatic cirrhosis with ascites and metabolic alkalosis, or those with resistant edema. Therefore, careful monitoring of serum electrolytes and BUN levels is important when using AA-AMILZIDE.

Non-melanoma skin cancer: An increased risk of non-melanoma skin cancer (NMSC) [basal cell carcinoma (BCC) and squamous cell carcinoma (SCC)] with increasing cumulative dose of hydrochlorothiazide exposure has been observed in two epidemiological studies based on the Danish National Cancer Registry. Photosensitizing actions of hydrochlorothiazide could act as a possible mechanism for NMSC. Patients taking hydrochlorothiazide should be informed of the risk of NMSC and advised to regularly check their skin for any new lesions and promptly report any suspicious skin lesions. Possible preventive measures such as limited exposure to sunlight and UV rays and, in case of exposure, adequate protection should be advised to the patients in order to minimize the risk of skin cancer. Suspicious skin lesions should be promptly examined potentially including histological examinations of biopsies. The use of hydrochlorothiazide may also need to be reconsidered in patients who have experienced previous NMSC.

Diabetes Mellitus: In diabetic patients, hyperkalemia has been commonly reported with the use of amiloride hydrochloride, particularly if they have chronic renal disease or pre-renal azotemia. Some deaths occurred in this last group of patients. Therefore, if therapy with amiloride hydrochloride is considered essential, the drug should be used with caution in diabetic or suspected diabetic patients and only after first determining the status of renal function. Careful monitoring of serum potassium levels is required throughout the therapy. One patient with poorly controlled diabetes mellitus who became severely hyperkalemic while on amiloride hydrochloride died following two repeated intravenous glucose tolerance tests. In diabetic patients, insulin requirements may be increased, decreased or unchanged due to the hydrochlorothiazide component. Diabetes mellitus which has been latent may become manifest during administration of thiazide diuretics.

Acute Respiratory Toxicity: Very rare severe cases of acute respiratory toxicity, including acute respiratory distress syndrome (ARDS) have been reported after taking hydrochlorothiazide. Pulmonary oedema typically develops within minutes to hours after hydrochlorothiazide intake. At the onset, symptoms include dyspnoea, fever, pulmonary deterioration and hypotension. If diagnosis of ARDS is suspected, AA-AMILZIDE should be withdrawn and appropriate treatment given. Hydrochlorothiazide should not be administered to patients who previously experienced ARDS following hydrochlorothiazide intake.

Metabolic or Respiratory Acidosis: Antikaliuretic therapy should be instituted only with caution in patients in whom respiratory or metabolic acidosis may occur, such as patients with cardiopulmonary disease or diabetes. If AA-AMILZIDE is given to these patients, frequent monitoring of acid base balance is necessary. Shifts in acid-base balance alter the ratio of extracellular/ intracellular potassium, and the development of acidosis may be associated with rapid increases in serum potassium levels.

Impaired Renal Function: Patients with impaired renal function other than those listed under CONTRA-INDICATIONS and who have BUN levels over 30mg per 100ml, serum creatinine levels over 1.5mg per 100ml, or blood urea values over 60mg per 100ml, should not receive the drug without careful, frequent monitoring of serum electrolyte, creatinine and BUN levels. Potassium retention associated with the use of AA-AMILZIDE is accentuated in the presence of renal impairment and may result in the rapid development of hyperkalemia. Prolongation of amiloride hydrochloride excretion was observed in patients with renal impairment. In patients with impaired renal function azotemia may be precipitated or increased by hydrochlorothiazide. Careful monitoring of such patients is therefore necessary.

Haematologic: Patients should be observed regularly for the possible occurrence of idiosyncratic reactions or blood dyscrasias.

Effects Related to Diuresis in Cirrhosis Patients: Patients with hepatic cirrhosis and ascites are intolerant of acute shifts in electrolyte balance and often have pre-existing hypokalemia as a result of associated secondary hyperaldosteronism. When oral diuretic therapy is used, these patients should be carefully monitored and diuresis should be gradual. Hepatic encephalopathy, manifested by tremors, confusion and coma has been reported in association with amiloride hydrochloride therapy. In cirrhotic patients receiving amiloride hydrochloride alone, jaundice associated with the underlying disease process has deepened in a few instances, but the relationship to the drug is uncertain. Patients should be observed regularly for the possible occurrence of liver dysfunction.

Immune: The possibility of exacerbation or activation of systemic lupus erythematosus has been reported with the thiazides.

Choroidal effusion, Acute Myopia and Secondary Angle-Closure Glaucoma: Hydrochlorothiazide, a sulfonamide, can cause an idiosyncratic reaction, resulting in choroidal effusion, acute transient myopia and/or secondary acute angle-closure glaucoma. Symptoms include acute onset of decreased visual acuity, blurred vision or ocular pain and typically occur within hours to weeks of drug initiation. Untreated acute angle-closure glaucoma can lead to permanent vision loss. The primary treatment is to discontinue hydrochlorothiazide as rapidly as possible. Prompt medical or surgical treatments may need to be considered if the intraocular pressure remains uncontrolled. Risk factors for developing acute angle-closure glaucoma may include a history of sulphonamide or penicillin allergy.

Peri-Operative Considerations: The antihypertensive effect of the drug may be enhanced in the post-sympathectomy patient.

Sensitivity/Resistance: Sensitivity reactions to thiazides may occur in patients with or without a history of allergy or bronchial asthma.

Photosensitivity: Photosensitivity reactions have been reported with the use of thiazide diuretics.

Other Precautions: Thiazides may decrease serum PBI levels without signs of thyroid disturbance. Calcium excretion is decreased by hydrochlorothiazide and magnesium excretion is increased. Pathological changes in the parathyroid glands with hypercalcemia and hypophosphatemia have been observed in a few patients on prolonged thiazide therapy. The common complications of hyperparathyroidism such as renal lithiasis, bone resorption and peptic ulceration have not been seen. Hyperuricemia may occur or gout may be precipitated.

Fertility: Reproduction studies in rats showed no evidence of impaired fertility.

Teratogenic Risk: Teratologic studies with amiloride hydrochloride in rabbits and mice revealed no evidence of harm to the foetus. At approximately 5 or more times the expected maximum daily dose for humans, some toxicity was seen in adult rats and rabbits and a decrease in rat pup growth and survival occurred.

Pregnancy: Because clinical experience is limited, AA-AMILZIDE is not recommended for use during pregnancy. In rats, a trace of drug crossed the placental barrier. Thiazides cross the placental barrier and appear in the cord blood. Therefore, the use of AA-AMILZIDE when pregnancy is present or suspected requires that the benefits of the drug be weighed against possible hazards to the fetus. These hazards include fetal or neonatal jaundice, thrombocytopenia and possibly other side effects that have occurred in the adult.

Breastfeeding: It is not known whether amiloride hydrochloride is excreted in human milk. In rats, secretion of amiloride hydrochloride in milk has been demonstrated. Thiazides appear in breast milk. Because of the potential for serious adverse reactions in nursing infants, if the use of AA-AMILZIDE is deemed essential, the patient should stop nursing.

Pediatric Use: The safety for use of amiloride hydrochloride in children has not been established: therefore, AA-AMILZIDE is not recommended in the pediatric age group.

Geriatric Use (>65 years of age): Evidence from clinical studies and experience suggests that use in the geriatric population is associated with differences in safety or effectiveness (See Serious Warnings and Precautions).

DRUG INTERACTIONS:

Serious Drug Interactions

Digoxin: Thiazide-induced electrolyte disturbances, i.e. hypokalemia, hypomagnesemia, increase the risk of digoxin toxicity, which may lead to fatal arrhythmic events. (See Drug-Drug Interactions, Digoxin)

Nonsteroidal anti-inflammatory drugs (NSAID): NSAID-related retention of sodium and water antagonises the diuretic and antihypertensive effects of thiazides. NSAID-induced inhibition of renal prostaglandins leading to decreases of renal blood flow, along with thiazide-induced decreases in GFR may lead to acute renal failure.

Patients with heart failure may be at particular risk (See Drug-Drug-Interactions, NSAID)
Antikaliuretic agents or Potassium Salts: Other antikaliuretic agents and potassium supplements are contraindicated in patients receiving AA-AMILZIDE (See Contraindications)

Drug-Behavioural Interactions: Consuming alcohol may modify the effect of this product. Avoid alcohol consumption during treatment.

Drug-Drug Interactions: The drugs listed in this table are based on either drug interaction case reports or studies, or potential interactions due to the expected magnitude and seriousness of the interaction (i.e., those identified as contraindicated).

Established or Potential Drug-Drug Interactions with Amiloride Hydrochloride

Proper/Common name	Source of Evidence	Effect	Clinical comment
Angiotensin-Converting Enzyme Inhibitors, Angiotensin II receptor antagonist	T	When amiloride hydrochloride is administered concomitantly with an angiotensin-converting enzyme inhibitor, an angiotensin II receptor antagonist, the risk of hyperkalemia may be increased.	If concomitant use of these agents is indicated because of demonstrated hypokalemia, they should be used with caution and with frequent monitoring of serum potassium.
Cyclosporine or Tacrolimus	T	When amiloride hydrochloride is administered concomitantly with cyclosporine or tacrolimus, the risk of hyperkalemia may be increased.	If concomitant use of these agents is indicated because of demonstrated hypokalemia, they should be used with caution and with frequent monitoring of serum potassium.
Lithium	T	Lithium should generally not be given with diuretics because they reduce the renal clearance of lithium and add a high risk of lithium toxicity.	
Nonsteroidal anti-inflammatory drugs (NSAID)	CT	In some patients, the administration of a non-steroidal anti-inflammatory agent can reduce the diuretic, natriuretic and hypertensive effects of diuretics. Concomitant administration of non-steroidal anti-inflammatory drugs (NSAIDs) and potassium-sparing agents, including amiloride hydrochloride, may cause hyperkalemia and renal failure, particularly in elderly patients.	When amiloride hydrochloride is used concomitantly with NSAIDs, renal function and serum potassium levels should be carefully monitored.

Established or Potential Drug-Drug Interactions with Hydrochlorothiazide

Proper/Common name	Source of Evidence	Effect	Clinical comment
Alcohol, barbiturates, or narcotics	C	Potential of orthostatic hypotension may occur.	Avoid alcohol, barbiturates or narcotics, especially with initiation of therapy.
Amphotericin B	T	Amphotericin B increases the risk of hypokalemia induced by thiazide diuretics.	Monitor serum potassium level.
Antidiabetic agents (e.g. insulin and oral	CT	Thiazide-induced hyperglycemia may compromise blood sugar control.	Monitor glycemic control, supplement potassium if

Proper/Common name	Source of Evidence	Effect	Clinical comment
hypoglycemic agents)		Depletion of serum potassium augments glucose intolerance.	necessary, to maintain appropriate serum potassium levels, and adjust diabetes medications as required.
Antihypertensive drugs	CT	Hydrochlorothiazide may potentiate the action of other antihypertensive drugs (e.g. guanethidine, methyldopa, beta-blockers, vasodilators, calcium channel blockers, ACEI, ARB, and direct renin inhibitors).	
Antineoplastic drugs, including cyclophosphamide and methotrexate	C	Concomitant use of thiazide diuretics may reduce renal excretion of cytotoxic agents and enhance their myelosuppressive effects	Hematological status should be closely monitored in patients receiving this combination. Dose adjustment of cytotoxic agents may be required.
Bile acid sequestrants, e.g. cholestyramine	CT	Bile acid sequestrants bind thiazide diuretics in the gut and impair gastrointestinal absorption by 43-85%. Administration of thiazide 4 hours after a bile acid sequestrant reduced absorption of hydrochlorothiazide by 30-35%.	Give thiazide 2-4 hours before or 6 hours after the bile acid sequestrant. Maintain a consistent sequence of administration. Monitor blood pressure, and increase dose of thiazide, if necessary.
Calcium and vitamin D supplements	C	Thiazides decrease renal excretion of calcium and increase calcium release from bone.	Monitor serum calcium, especially with concomitant use of high doses of calcium supplements. Dose reduction or withdrawal of calcium and/or vitamin D supplements may be necessary.
Carbamazepine	C	Carbamazepine may cause clinically significant hyponatremia. Concomitant use with thiazide diuretics may potentiate hyponatremia.	Monitor serum sodium levels. Use with caution.
Corticosteroids, and adrenocorticotrophic hormone (ACTH)	T	Intensified electrolyte depletion, particularly hypokalemia, may occur.	Monitor serum potassium, and adjust medications, as required.
Digoxin	CT	Thiazide-induced electrolyte disturbances, i.e. hypokalemia, hypomagnesemia, increase the risk of digoxin toxicity, which may lead to fatal arrhythmic events.	Concomitant administration of hydrochlorothiazide and digoxin requires caution. Monitor electrolytes and digoxin levels closely. Supplement potassium or adjust doses of digoxin or thiazide, as required.
Drugs that alter GI motility, i.e., anti-cholinergic agents, such as atropine and prokinetic agents, such as metoclopramide, domperidone	CT, T	Bioavailability of thiazide diuretics may be increased by anticholinergic agents due to a decrease in gastrointestinal motility and gastric emptying. Conversely, prokinetic drugs may decrease the bioavailability of thiazide diuretics.	Dosage adjustment of thiazide may be required
Gout medications	T, RCS	Thiazide-induced hyperuricemia may	Dosage adjustment of gout

Proper/Common name	Source of Evidence	Effect	Clinical comment
(allopurinol, uricosurics, xanthine oxidase inhibitors)		compromise control of gout by allopurinol and probenecid. The co-administration of hydrochlorothiazide and allopurinol may increase the incidence of hypersensitivity reactions to allopurinol.	medications may be required.
Lithium	CT	Thiazide diuretics reduce the renal clearance of lithium and add a high risk of lithium toxicity.	Concomitant use of thiazide diuretics with lithium is generally not recommended. If such use is deemed necessary, reduce lithium dose by 50% and monitor lithium levels closely.
Nonsteroidal anti-inflammatory drugs (NSAID)	CT	NSAID-related retention of sodium and water antagonises the diuretic and antihypertensive effects of thiazides. NSAID-induced inhibition of renal prostaglandins leading to decreases of renal blood flow, along with thiazide-induced decreases in GFR may lead to acute renal failure. Patients with heart failure may be at particular risk.	If combination use is necessary, monitor renal function, serum potassium, and blood pressure closely. Dose adjustments may be required.
Selective serotonin reuptake inhibitors (SSRIs, e.g. citalopram, escitalopram, sertraline)	T, C	Concomitant use with thiazide diuretics may potentiate hyponatremia.	Monitor serum sodium levels. Use with caution.
Skeletal muscle relaxants of the curare family, e.g., tubocurare	C	Thiazide drugs may increase the responsiveness of some skeletal muscle relaxants, such as curare derivatives	
Topiramate	CT	Additive hypokalemia. Possible thiazide-induced increase in topiramate serum concentrations.	Monitor serum potassium and topiramate levels. Use potassium supplements, or adjust topiramate dose as necessary.

Legend: C= Case Study; RCS =Retrospective Cohort Study; CT = Clinical Trial; T =Theoretical

ADVERSE REACTIONS: While rare, the most serious adverse effect of AA-AMILZIDE (hydrochlorothiazide and amiloride hydrochloride) is symptomatic hyperkalemia. Other metabolic changes that occur are asymptomatic hyperkalemia, hypokalemia and hyponatremia.

The following incidence of other adverse reactions was reported in patients treated with a combination of hydrochlorothiazide and amiloride hydrochloride.

	Common Incidence $\geq 3\%$	Common Incidence $>1\%$ - $<3\%$	Uncommon Incidence $\leq 1\%$
Cardiac disorders (In 4.3% of patients)		Arrhythmia	Tachycardia Angina pectoris
Eye disorders (In 13.9% of patients)			Visual disturbance
Gastrointestinal disorders (In 7.1% of patients)	Nausea (3.7%)	Diarrhea Gastrointestinal pain.	Constipation GI bleeding

	Common Incidence $\geq 3\%$	Common Incidence $>1\%$ - $<3\%$	Uncommon Incidence $\leq 1\%$
		Abdominal pain.	GI disturbance Appetite changes Abdominal fullness Hiccups Vomiting Flatulence Bad taste
General disorders and administration site conditions	Weakness (4.0%) (13.9% of patients)	Fatigue/tiredness (In 2.6% of patients)	Malaise (In 2.6% of patients) Chest pain (In 3.7% of patients) Thirst (In 7.1% of patients)
Injury poisoning and procedural complications (In 4.3% of patients)			Digitalis Toxicity
Metabolism and Nutrition Disorders (In 0.9% of patients)	Anorexia (3.7%) (In 7.1% of patients)		Gout Dehydration
Musculoskeletal and connective tissue disorders (In 3.7% of patients)		Leg ache	Muscle cramps/ spasm Joint pain Back pain
Nervous system disorders (In 13.9% of patients)	Headache (7.8%) Dizziness (6.1%)		Paresthesia Numbness Stupor Vertigo
Psychiatric disorders (In 13.9% of patients)			Insomnia Nervousness Depression Sleepiness Mental confusion
Renal and Urinary disorders (In 1.7% of patients)			Nocturia Dysuria Incontinence
Reproductive system and breast disorders (In 1.7% of patients)			Impotence
Respiratory, thoracic and mediastinal disorders (In 2.6% of patients)		Dyspnea	Nasal congestion
Skin and subcutaneous tissue disorders (In 5.2% of patients)	Rash (3.4%)	Pruritus	Flushing
Vascular disorders (In 4.3% of patients)			Orthostatic hypotension

Other adverse reactions that have been reported with the individual components are listed below:

Blood and lymphatic system disorders: Agranulocytosis, aplastic anemia, hemolytic anemia, leukopenia, purpura, thrombocytopenia, neutropenia.

Eye disorders: Transient blurred vision, xanthopsia.

Immune system disorders: Anaphylactic reactions.

Gastrointestinal disorders: Activation of pre-existing peptic ulcer, cramping, gastric irritation, pancreatitis, dry mouth, sialadenitis.

General disorders and administration site conditions: Fever
Hepatobiliary disorders: Abnormal liver function, jaundice (intrahepatic cholestatic jaundice).
Metabolism and nutrition disorders: Hyperglycemia, hyperuricemia.
Psychiatric disorders: Restlessness.
Renal and urinary disorders: Glycosuria.
Respiratory, thoracic and mediastinal disorders: Respiratory distress including pneumonitis
Skin and subcutaneous tissue disorders: Photosensitivity, urticaria.
Vascular disorders: Necrotizing angitis (vasculitis, cutaneous vasculitis).

Post-Market Adverse Reactions:

Eye disorder: Choroidal effusion, acute myopia, acute angle-closure glaucoma (unknown frequency)
Neoplasms benign, malignant and unspecified (incl cysts and polyps)
Frequency 'not known': Non-melanoma skin cancer (Basal cell carcinoma and Squamous cell carcinoma)
Based on available data from epidemiological studies, cumulative dose-dependent association between hydrochlorothiazide and NMSC has been observed.
Respiratory, thoracic and mediastinal disorders
Frequency 'very rare': Acute respiratory distress syndrome (ARDS)

SYMPTOMS AND TREATMENT OF OVERDOSAGE: No data are available in regard to overdosage in humans with AA-AMILZIDE (hydrochlorothiazide and amiloride hydrochloride) or with the amiloride hydrochloride component. The most common signs and symptoms to be expected from overdosage with AA-AMILZIDE are dehydration and electrolyte imbalance. Serum electrolytes should be carefully monitored with special attention to potassium levels. Cardiac arrhythmias may be caused by abnormal potassium levels. Digitalized patients are especially prone to arrhythmias. No specific information is available on the treatment of AA-AMILZIDE overdosage and no specific antidote is available. Treatment is symptomatic and supportive. Therapy with AA-AMILZIDE should be discontinued and the patient observed closely. Suggested measures include induction of emesis and/or gastric lavage. It is not known whether the drug is dialyzable.

MODE OF ADMINISTRATION: Oral

DOSAGE AND ADMINISTRATION: Optimal dosage should be established by the individual titration of the components. Maintenance doses may be lower than those required to initiate diuresis; therefore, reduction in the daily dosage should be attempted when the patient's weight is stabilized. In cirrhotic patients, gradual weight reduction is especially desirable to reduce the likelihood of untoward reactions associated with diuretic therapy.

Hepatic Cirrhosis with Ascites and Edema: The usual maintenance dose of AA-AMILZIDE (hydrochlorothiazide and amiloride hydrochloride) is 1 tablet given once a day. The dosage should not exceed 4 tablets a day in single or divided doses.

Edema of Cardiac Origin: The usual maintenance dose of AA-AMILZIDE is 1 or 2 tablets given once a day in divided doses. The dosage should not exceed 4 tablets a day. Therapy may be on an intermittent basis.

Hypertension: The usual maintenance dosage is 1 or 2 tablets given once a day or in divided doses. The dosage should not exceed 4 tablets a day.

AVAILABILITY: AA-AMILZIDE Tablets, containing 50mg hydrochlorothiazide and 5mg amiloride hydrochloride, are peach coloured, diamond shaped tablets, scored and identified "APO" above the score, "5/50" below the score. Available in PVC Blisters of 500 (50x10's) & 1000 (100 x 10's) tablets.

STORAGE CONDITIONS: Store below 30°C.

MANUFACTURER: Apotex Inc, 150 Signet Drive, Weston (Toronto), Ontario, Canada, M9L 1T9.

DATE OF REVISION OF PACKAGE INSERT: 5th March 2026

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ST0090C-4